

PreHOI-Rank Manuscript Draft Export

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Title Page

Title Page Draft

Status: five-author metadata filled. Student co-author consent, CRediT roles, and final author-name spelling are confirmed. Journal-system checks remain pending.

Full Title

PreHOI-Rank: Affordance-Grounded Candidate Ranking for Pre-Contact 3D Hand-Object Interaction Forecasting

Short Title

PreHOI-Rank for Pre-Contact Hand-Object Forecasting

Authors

Order	Full author name	Affiliation number	Email	ORCID
1	Al Jubair Hossain	1	jubair.hossain@aiub.edu	0009-0005-3498-5826
2	ASIF SHAHRIAR SIAM	1	On file internally	Not provided
3	MD. ABUBOKOR SIDDIK ROJAN	1	On file internally	Not provided
4	Maria Sultana Alif	1	On file internally	Not provided
5	Seyam Rahman Nayem	1	On file internally	Not provided

Affiliation

Number	Institution	Address
1	American International University-Bangladesh (AIUB)	Kuratoli, Khilkhet, Dhaka 1229, Bangladesh

Corresponding Author

- Name: Al Jubair Hossain
- Email: jubair.hossain@aiub.edu
- Affiliation: American International University-Bangladesh (AIUB)

- Address: Kuratoli, Khilkhet, Dhaka 1229, Bangladesh

CRedit Author Contributions

Student co-author contribution roles have been confirmed from the provided consent responses.

- Al Jubair Hossain: Conceptualization, Methodology, Software, Validation,

Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review and editing, Visualization, Project administration, Supervision.

- ASIF SHAHRIAR SIAM: Investigation, Data curation, Validation, Visualization,

Writing - review and editing.

- MD. ABUBOKOR SIDDIK ROJAN: Software support, Validation, Formal analysis

support, Data curation, Writing - review and editing.

- Maria Sultana Alif: Literature review support, Data curation, Investigation,

Visualization support, Writing - review and editing.

- Seyam Rahman Nayem: Dataset preparation support, Reproducibility checks,

Validation, Documentation support, Writing - review and editing.

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using filled AIUB/Bangladesh affiliation.

- ☒ Corresponding author details added.

Main Manuscript

PreHOI-Rank: Affordance-Grounded Candidate Ranking for Pre-Contact 3D Hand-Object Interaction Forecasting

Author Information

Order	Author	Affiliation(s)	Email	ORCID
1	Al Jubair Hossain	1	jubair.hossain@aiub.edu	0009-0005-3498-5826
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Affiliations:

- [1] American International University-Bangladesh (AIUB), Kuratoli, Khilkhhet, Dhaka 1229, Bangladesh

Corresponding Author

- Name: Al Jubair Hossain
- Email: jubair.hossain@aiub.edu
- Affiliation: American International University-Bangladesh (AIUB)

Abstract

Pre-contact hand-object forecasting asks whether an egocentric perception system can infer the likely object of a future hand interaction before contact occurs. The problem is relevant to assistive augmented reality, wearable interfaces, and robotic systems that must reason from partial observations of human intent. We present PreHOI-Rank, a leakage-safe candidate-ranking formulation for pre-contact 3D hand-object interaction forecasting. Rather than predicting a target from a global closed-set vocabulary, PreHOI-Rank scores the visible object candidates in an observation window and jointly predicts a future hand-pose representation. We construct derived affordance-grounded target-object proxy labels from HOT3D-Clips [hot3d2025], [hot3dclips2026] using forecast-frame hand-object proximity, while preventing forecast-frame images, annotations, candidate scores, and metadata from entering the model input. Stable candidate ordering is enforced to avoid candidate-position leakage. On a local 50-clip HOT3D-Clips subset with clip-level splits, the order-safe non-vision-language candidate ranker achieves five-seed paper-candidate diagnostics of 0.7499 $\pm$ 0.0450 Top-1 candidate accuracy, 0.9699 $\pm$ 0.0161 Top-3 candidate accuracy, 0.8605 $\pm$ 0.0221 MRR, and 0.4102 $\pm$ 0.0051 future pose-vector MAE. These results support candidate ranking as a practical formulation for pre-contact target forecasting under derived labels, while leaving important limitations around the local 50-clip subset, residual class imbalance, proxy-label

assumptions, and pose-vector rather than MPJPE-style evaluation.

Keywords

Egocentric video; hand-object interaction; pre-contact forecasting; candidate ranking; HOT3D-Clips; affordance proxy labels; leakage prevention; 3D hand pose.

Highlights

- PreHOI-Rank frames pre-contact hand-object forecasting as candidate-level ranking over visible objects.
- Derived affordance-grounded proxy labels are constructed from forecast-frame hand-object proximity while forecast-frame features are excluded from model inputs.
- The evaluation protocol uses clip-level splits, stable candidate ordering, and position baselines to reduce temporal and candidate-order leakage.
- The 50-clip HOT3D-Clips local subset is used as the primary controlled result, with a 75-clip expansion reported as a robustness/scalability check.
- Pose metrics are reported as MANO/UmeTrack pose-parameter vector MAE/MSE; MPJPE is not reported.

1. Introduction

Before contact occurs, human object interactions are often foreshadowed by hand motion, the body-centered viewpoint, and the changing spatial relation between hands and nearby objects. Anticipating this interaction can help egocentric systems act earlier: an assistive AR interface may prepare object-specific guidance, a robot may infer a user's likely target, and a wearable perception system may forecast hand-object contact before it is visually obvious.

This paper focuses on pre-contact target-object forecasting in egocentric video. The goal is to use an observation window before contact to infer which visible object candidate is most likely to become the future hand-object target, while also predicting a future hand-pose representation. Figure 1 summarizes this problem setting.

[Figure 1 here: paper/figures/fig1_problem_overview.png; PDF version: paper/figures/fig1_problem_overview.pdf.]

A direct approach would cast the task as global object classification. Although convenient, that framing is not always well matched to egocentric interaction. In many frames, the practical question is not "which object class in the dataset is present?" but "which currently visible object will the hand approach?" Global classification can also make evaluation sensitive to local class coverage and can hide candidate-order or temporal leakage effects.

We therefore propose PreHOI-Rank: a candidate-level formulation that ranks the visible object candidates in each observation window. The current strongest model is intentionally simple. It uses observation-window hand/object metadata and object candidate features, excludes forecast-frame inputs, sorts candidates with stable identifiers, and trains with a candidate-ranking objective plus future pose-vector regression. More complex vision-language and transformer variants are retained as exploratory ablations, but they are not the main claim because they have not yet outperformed the order-safe non-vision-language ranker under repeated-seed evaluation.

This paper makes four contributions:

1. We formulate pre-contact target-object forecasting as ranking visible object

candidates rather than closed-set global classification.

2. We define a reproducible derived target-object proxy from forecast-frame hand-object proximity and clearly distinguish it from human contact or action annotations.

3. We introduce a leakage- and order-bias-safe evaluation protocol with observation-frame-only inputs, clip-level splitting, stable candidate ordering, and position-only baselines.

4. We provide five-seed paper-candidate diagnostics on a 50-clip local HOT3D-Clips subset, showing that the order-safe candidate ranker is stronger than candidate-position baselines and improves over the earlier 25-clip pilot.

The manuscript is therefore framed as a reproducible method and evaluation protocol for pre-contact candidate ranking with derived affordance-grounded proxy labels, not as a benchmark-superiority claim.

2. Related Work

Research on 3D hand-object interaction has produced increasingly rich datasets for hand pose, object pose, and hand-object state estimation. HO-3D and HOnnotate support 3D hand-object pose estimation in object-interaction scenes [ho3d2020], while DexYCB provides multi-view RGB-D data for hand-object grasping and pose estimation [dexycb2021]. HOT3D extends this line of work to egocentric multi-view video with 3D hand/object tracking, object models, and multiple annotation streams [hot3d2025]. The present work builds on HOT3D-Clips [hot3dclips2026], while using only a local 50-clip subset and without redistributing the dataset.

Hand-pose representations are also central to this work. MANO provides a widely used parametric hand model [mano2017], and HOT3D includes hand annotations compatible with MANO/UmeTrack-style representations [umetrack2022]. The current experiments regress future pose vectors directly. They do not yet convert the pose representation to 3D joints for MPJPE-style evaluation, which remains an important future improvement.

Egocentric interaction datasets and action-anticipation benchmarks motivate the pre-contact setting. First-person action and hand-action datasets such as FPHA [fpha2018], EPIC-KITCHENS [epickitchens2020], and AssemblyHands [assemblyhands2023] show the value of modeling hands, objects, and temporal context in first-person video. Our focus is narrower: we rank visible object candidates before contact, using derived proxy labels when direct target-object/contact annotations are unavailable.

Affordance and contact reasoning are closely related because the target proxy uses hand-object proximity at a future frame. However, the proxy should not be interpreted as an official semantic affordance annotation or human contact annotation. ContactPose provides a related contact-focused reference for hand-object grasp/contact modeling [contactpose2020].

The candidate-ranking formulation is related to learning-to-rank and object-centric interaction prediction, where the model scores a set of candidates rather than predicting a single global class. This design is related to learning-to-rank methods such as RankNet [burges2005ranknet]. In this paper, that formulation is also a safety choice: ranking visible candidates makes candidate-order leakage measurable and allows position-only baselines.

Temporal leakage is a central concern in forecasting. A method that uses forecast-frame features as input can appear effective while violating the pre-contact premise. We therefore enforce observation-frame-only inputs and clip-level train/validation/test splitting, following the broader machine-learning principle that target information must not be available through features or evaluation design at prediction time

[@kaufman2012leakage].

3. Methods

3.1 HOT3D-Clips Local Subset

We use HOT3D-Clips as the current data source [@hot3d2025], [@hot3dclips2026]. HOT3D-Clips provides curated HOT3D subsequences in WebDataset shard format, with image streams and per-frame annotations for hands, objects, cameras, and metadata. The primary controlled experiment uses a local subset of 50 downloaded shards, not the complete HOT3D or HOT3D-Clips release. A later 75-clip expansion is reported as a robustness/scalability analysis rather than as the main result.

The local subset contains 6500 proxy-labeled samples before optimized class-based filtering. After filtering and optimized clip-level splitting, the final protocol uses 4175 training samples, 1040 validation samples, and 910 test samples. The split uses 35 training clips, 8 validation clips, and 7 test clips. Clip-level splitting is required because neighboring samples from the same clip can be highly correlated.

3.2 Sample Construction

Each sample is a pre-contact forecasting window with 16 observation frames and a forecast frame 5 frames after the observation window. If the observation frames are indexed from t to $t + 15$, the forecast target is defined at $t + 20$. Model inputs may use only the observation frames. The forecast frame is used only to derive the target-object proxy and the future hand-pose vector.

Each sample stores:

- observation frame identifiers;
- the forecast frame identifier;
- observation-frame hand/object metadata;
- observation-frame object candidates;
- the forecast-frame derived target-object proxy label;
- the future MANO/UmeTrack pose vector;
- metadata including clip ID and safety flags.

The key safety flag is `input_uses_forecast_frame=false`. Any run that violates this flag should be excluded.

3.3 Ethics and Data Use

This study uses an existing third-party dataset. The authors did not collect new human-subject data. HOT3D-Clips data, image streams, raw annotations, WebDataset shards, object models, and other restricted dataset files are not redistributed by this project. Users must obtain HOT3D-Clips from the official provider and follow the dataset license, access conditions, and citation requirements.

The current manuscript should include a final dataset license/access note after the official HOT3D-Clips terms are rechecked. The repository should contain code and regeneration scripts, not downloaded dataset files.

3.4 Derived Target-Object Proxy Label

The current target-object labels are derived proxy labels, not direct HOT3D official annotations. This distinction is central to the paper.

For each forecast frame, the proxy generator collects visible hand boxes and visible object boxes. It forms a hand union box and scores each candidate object using overlap with the hand union box and normalized center distance. The highest-scoring forecast-frame object is selected as the target-object proxy. Figure 2 illustrates this proxy-label construction.

[Figure 2 here: paper/figures/fig2_proxy_label_generation.png; PDF version: paper/figures/fig2_proxy_label_generation.pdf.]

The proxy is intended to approximate the object most aligned with future hand contact or affordance use. It is not a human action label, not an official contact annotation, and not a semantic intention label. It can be wrong when the nearest future object is not the intended object, when bounding boxes are noisy, when multiple objects are close to the hand, or when the action depends on context that is not captured by proximity.

The proxy is allowed only as a supervised target. The following forecast-frame signals are not allowed as model inputs:

- forecast-frame images;
- forecast-frame object boxes;
- forecast-frame hand boxes;
- forecast-frame proxy scores;
- forecast-frame candidate ordering;
- forecast-frame metadata.

This separation preserves the pre-contact forecasting setting: the model sees only the past observation window and predicts a future target proxy.

3.5 Candidate-Ranking Formulation

Given an observation window and a set of visible object candidates, PreHOI-Rank predicts a score for each candidate. The target is the candidate matching the derived forecast-frame target-object proxy. Candidates are padded to a fixed maximum count and accompanied by a candidate mask so invalid padded candidates do not contribute to the ranking loss.

This formulation differs from global object classification. The model is not asked to choose among every object class in the dataset. Instead, it chooses among the objects visible in the current observation context. This better matches egocentric interaction forecasting, where the target is constrained by what is present around the hand.

3.6 Input Features

The current strongest model uses observation-window metadata and object candidate features. Temporal metadata summarizes hand/object context over the 16 observation frames. Candidate features describe each visible object from the observation window, including geometry, visibility, hand-object proximity computed from observation frames, and object identity features. No image, CLIP, or forecast-frame features are used in the final candidate-ranker protocol.

The model also predicts a future MANO/UmeTrack pose vector. This auxiliary target keeps the formulation connected to 3D hand-object forecasting, although future work should convert pose parameters to 3D joints for MPJPE-style evaluation.

3.7 Model Architecture

The non-vision-language candidate ranker contains:

- a temporal encoder for observation-window metadata;
- an object-candidate encoder for padded candidate features;
- a fusion module that combines temporal context with candidate features;
- a candidate score head producing one score per candidate;
- a future pose-vector regression head.

The training objective combines candidate-ranking cross-entropy with pose regression loss. Figure 3 shows the current architecture at a high level.

[Figure 3 here: [paper/figures/fig3_prehoi_rank_architecture.png](#); PDF version: [paper/figures/fig3_prehoi_rank_architecture.pdf](#).]

3.8 Candidate-Order Safety

Candidate order can create a hidden leakage path. Earlier pilot checks showed that unsafe ordering could place the target candidate near position 0 too often, inflating candidate-ranking metrics. PreHOI-Rank therefore uses `candidate_order: stable_uid`, which sorts candidates by stable object identifier or object name. The model is not evaluated with raw/as-is ordering, proxy-score ordering, target-aware ordering, or any order based on forecast-frame information.

The evaluation also reports position-only baselines, including candidate-0 Top-1, first-3 Top-3, and position-only MRR. A valid ranker must be interpreted relative to these baselines.

4. Experimental Setup

The final-protocol candidate-ranker experiment uses the 50-clip local HOT3D-Clips subset and optimized clip-level splits. The protocol is summarized in Table 2 and Figure 4.

[Figure 4 here: [paper/figures/fig4_protocol_safety.png](#); PDF version: [paper/figures/fig4_protocol_safety.pdf](#).]

[Table 2 here: [paper/tables/protocol_safety_table.md](#).]

The required safety rules are:

- use only observation-frame inputs;
- keep `input_uses_forecast_frame=false`;
- use forecast-frame proxy labels only as supervised targets;
- use `candidate_order: stable_uid`;
- exclude raw/as-is, proxy-sorted, and target-aware candidate orderings;
- split train/validation/test by clip ID, not random sample ID;
- call labels derived proxy labels, not official annotations.

The final candidate-ranker protocol uses five seeds: 42, 123, 2026, 7, and 99. Metrics are reported as mean +/- standard deviation over seeds.

The ranking metrics are:

- Top-1 candidate accuracy;

- Top-3 candidate accuracy;
- mean reciprocal rank (MRR).

The pose metrics are:

- future pose-vector mean squared error (MSE);
- future pose-vector mean absolute error (MAE).

These pose metrics are computed on the current pose-vector representation, not on reconstructed 3D joints.

5. Results

5.1 Main 50-Clip Five-Seed Result

Table 1 reports the main paper-candidate result for the order-safe non-vision-language candidate ranker on the 50-clip local HOT3D-Clips subset.

[Table 1 here: [paper/tables/results_table_prehoi_rank.md](#).]

Metric	Mean +/- Std
Top-1 candidate accuracy	0.7499 +/- 0.0450
Top-3 candidate accuracy	0.9699 +/- 0.0161
MRR	0.8605 +/- 0.0221
Pose MSE	0.4301 +/- 0.0116
Pose MAE	0.4102 +/- 0.0051

The candidate ranker substantially exceeds the test-split candidate-0 and position-only MRR baselines under stable UID ordering. Top-3 accuracy, however, requires a more cautious reading because some samples have small candidate sets. For that reason, first-3 and random Top-3 baselines are reported alongside model metrics.

5.2 25-Clip Versus 50-Clip Comparison

The 50-clip protocol improves over the earlier 25-clip pilot:

Setting	Top-1	Top-3	MRR	Pose MAE
25-clip, 3-seed pilot	0.5624	not reported	0.7502	0.4412
50-clip, 5-seed primary protocol	0.7499 +/- 0.0450	0.9699 +/- 0.0161	0.8605 +/- 0.0221	0.4102 +/- 0.0051

Figure 5 visualizes this comparison together with the 75-clip robustness analysis described below.

[Figure 5 here: [paper/figures/fig5_25clip_vs_50clip_results.png](#); PDF version: [paper/figures/fig5_25clip_vs_50clip_results.pdf](#).]

The improvement suggests that data expansion and split quality materially affect this task. It does not establish final generalization to the complete HOT3D release.

5.3 Robustness Analysis on a 75-Clip Expansion

We also expanded the local HOT3D-Clips subset to 75 shards and reran the same five-seed candidate-ranker protocol. The expansion increased the proxy sample count from 6500 to 9750 and increased the optimized eligible proxy classes from 23 to 30. However, the resulting split was broader and harder: validation and test still missed some eligible classes, and the number of classes shared across train, validation, and test decreased from 20 in the 50-clip protocol to 17 in the 75-clip protocol.

Setting	Top-1	Top-3	MRR	Pose MAE
50-clip, primary protocol	0.7499 +/- 0.0450	0.9699 +/- 0.0161	0.8605 +/- 0.0221	0.4102 +/- 0.0051
75-clip, robustness protocol	0.7115 +/- 0.0571	0.9789 +/- 0.0009	0.8340 +/- 0.0343	0.4676 +/- 0.0096

The 75-clip protocol maintained high Top-3 performance and had a lower candidate-0 position baseline on the test split. At the same time, Top-1, MRR, and pose-vector MAE were weaker than in the cleaner 50-clip protocol. We therefore use the 50-clip protocol as the primary controlled evaluation and report the 75-clip protocol as a robustness/scalability analysis under a larger but less balanced local subset.

5.4 Exploratory Ablations

During development, metadata-only, object-aware, image-statistics, frozen-CLIP, candidate-ranker, vision-language candidate-ranker, PreHOI-Former v1, and PreHOI-Former v2 variants were tested as pilot experiments. Some early runs were excluded because of leakage or candidate-order safety concerns.

The current manuscript should treat vision-language and PreHOI-Former variants as exploratory ablations. They are useful for understanding the design space, but the repeated-seed evidence currently favors the simpler order-safe non-vision-language candidate ranker.

6. Discussion

The results support candidate ranking as the most evidence-backed formulation in the current project. The model benefits from asking a local, interaction-specific question: among the visible candidates, which object is most likely to become the future hand-object target? This framing is closer to the egocentric interaction problem than predicting a global object class.

The safety protocol is as important as the model itself. Because the target proxy is derived from the forecast frame, any forecast-frame feature in the input would make the task artificially easy. Candidate ordering can create a similar problem when candidates are sorted by target-aware quantities. The manuscript therefore reports explicit leakage and order-bias controls rather than interpreting candidate-ranking metrics in isolation.

The finding that the non-vision-language ranker outperforms current vision-language variants is also informative. It suggests that hand-object geometry and candidate-level structure are strong signals for this proxy task. Vision-language fusion may still be valuable, but it should be redesigned and evaluated as an ablation under the same protocol instead of being claimed as the central contribution.

7. Limitations and Threats to Validity

Derived Proxy Labels

The target-object labels are derived from forecast-frame hand-object proximity. They are not human-annotated action labels, direct contact annotations, or official HOT3D target-object annotations. The

proxy may fail when interaction intent differs from closest-object geometry.

Local 50-Clip Subset

The primary result uses 50 local HOT3D-Clips shards, not the complete dataset. A 75-clip robustness run was also completed, but it introduced a harder and less balanced split and did not replace the 50-clip result as the primary controlled evaluation. Full-dataset or larger-subset evaluation is still needed before broad generalization claims.

Residual Class Imbalance

The optimized split improves class coverage, but some imbalance remains. The test split is missing food_waffles, potato_masher, and spatula_red, and the train split has low counts for bottle_ranch, cellphone, and mug_white. These warnings should remain visible in the final manuscript.

Pose Metric Limitation

Pose evaluation currently uses MANO/UmeTrack pose-vector MSE and MAE. These metrics are useful for pipeline validation, but they are not equivalent to 3D joint MPJPE. Future work should convert pose representations to 3D joints if a validated conversion path is available.

Candidate Visibility Assumption

The candidate-ranking task assumes that the relevant object is represented in the visible candidate set. If the future target is absent, occluded, incorrectly detected, or missing from the candidate annotations, the ranking formulation cannot select it correctly.

Exploratory Model Scope

Vision-language and PreHOI-Former variants remain exploratory. Current evidence does not justify claiming that those components are the main performance driver.

8. Data and Code Availability

HOT3D-Clips is a third-party dataset distributed by its original providers. The authors do not redistribute raw HOT3D-Clips data, WebDataset shards, image streams, object models, or restricted annotations. Readers should obtain the dataset directly from the official provider and follow the provider's license, access, and citation requirements.

The code is available in the PreHOI-Rank repository at <https://github.com/XessX/prehoi-rank-hot3d>; The final/current v0.1.2 Zenodo archive DOI is 10.5281/zenodo.20736962. The repository should include scripts for HOT3D-Clips inspection, proxy-label generation, split optimization, leakage checks, candidate-order bias checks, training, metric collection, and figure generation. Generated sample indexes can be regenerated from downloaded HOT3D-Clips shards by authorized users. Whether derived index files can be shared directly must be confirmed against the dataset terms before submission.

The authors did not collect new private human-subject data. Trained checkpoints, logs, and derived summaries should be shared only if allowed by the dataset terms and only if they do not reveal restricted dataset content.

9. Ethics and Data Use

The authors did not collect new human-subject data for this study. Experiments use HOT3D-Clips, a third-party dataset distributed by its original providers. The raw dataset, video frames, WebDataset shards, object models, annotations, and restricted dataset files are not redistributed in this submission package.

Users must obtain HOT3D-Clips through the official access route and comply with the dataset license, access conditions, citation requirements, and data-use terms. The derived proxy-label protocol and sample-index generation scripts are intended to be reproducible by authorized users with their own downloaded copy of the dataset.

10. Conflict of Interest

The authors declare no competing interests.

11. Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

12. Acknowledgments

No specific funding was received for this work. No additional acknowledgments were provided.

13. Conclusion

PreHOI-Rank frames pre-contact hand-object target forecasting as candidate ranking over visible objects. The formulation combines derived affordance-grounded proxy labels, observation-frame-only inputs, stable candidate ordering, and clip-level splitting to reduce common leakage paths in temporal forecasting. On the current 50-clip local HOT3D-Clips subset, the order-safe candidate ranker produces stable five-seed paper-candidate diagnostics and improves over the earlier 25-clip pilot. The work is not a complete-dataset benchmark, but it provides a reproducible and cautious foundation for pre-contact hand-object candidate ranking.

14. References Placeholder Note

References should be generated from references.bib after final citation and journal-style checks. The current draft uses citation-style placeholders such as [@hot3d2025] for Markdown/Pandoc-style processing.

Figure Captions

Figure Caption List

Status: draft captions for the formatted submission package.

Figure 1

Figure 1. Pre-contact candidate-ranking setup. Given an observation window, PreHOI-Rank ranks visible object candidates as likely future hand-object targets. The forecast frame is used only to define a derived proxy target and is not used as model input.

Files:

- fig1_problem_overview.png
- fig1_problem_overview.pdf

Figure 2

Figure 2. Derived target-object proxy construction. The proxy target is selected at the forecast frame using hand-object box overlap and normalized center distance. These forecast-frame quantities define the supervised target only and are excluded from model inputs.

Files:

- fig2_proxy_label_generation.png
- fig2_proxy_label_generation.pdf

Figure 3

Figure 3. PreHOI-Rank model overview. The model encodes observation-window metadata and object candidates, ranks visible candidates with a masked scoring head, and regresses a future hand-pose vector. The current strongest model does not require vision-language features.

Files:

- fig3_prehoi_rank_architecture.png
- fig3_prehoi_rank_architecture.pdf

Figure 4

Figure 4. Evaluation safety protocol. Valid runs use clip-level splits, observation-frame inputs, stable UID candidate ordering, and position baselines. Runs with forecast-frame inputs or target-aware candidate ordering are excluded.

Files:

- fig4_protocol_safety.png
- fig4_protocol_safety.pdf

Figure 5

Figure 5. Candidate-ranker protocol diagnostics across 25, 50, and 75 local HOT3D-Clips subsets. Expanding from 25 to 50 local shards improves Top-1 candidate accuracy, MRR, and pose-vector MAE

under the same order-safe candidate-ranking formulation. The 75-clip robustness split increases data scale and class diversity but is harder and less balanced; it maintains high Top-3 performance while Top-1, MRR, and pose-vector MAE are weaker than the 50-clip primary protocol. Bars show mean values with seed-standard-deviation error bars where available. Lower is better for pose-vector MAE.

Files:

- fig5_25clip_vs_50clip_results.png
- fig5_25clip_vs_50clip_results.pdf

Table Captions

Table Caption List

Status: draft captions for the formatted submission package.

Table 1

Table 1. PreHOI-Rank candidate-ranker diagnostics on local HOT3D-Clips subsets. The table reports the 50-clip five-seed primary result, per-seed diagnostics, a 25/50/75 clip scale comparison, and candidate-position baselines. The 75-clip setting is reported as a robustness/scalability check rather than replacing the 50-clip primary result.

Source file:

- tables/results_table_prehoi_rank.md

Table 2

Table 2. Protocol safety checks for leakage-safe candidate ranking. The table summarizes forecast-frame input exclusion, target-proxy use, stable UID candidate ordering, clip-level splitting, candidate-order bias diagnostics, and remaining split warnings.

Source file:

- tables/protocol_safety_table.md